Zassenhaus Lemma or butterfly lemma:

Let $H, K$ be subgroups of a group $G$. And let $H^*$ and $K^*$ be normal groups of $H$ and $K$. Then we have

(1) $H^*(H\cap K^*) \trianglelefteq H^*(H\cap K)$

(2) $K^*(H^*\cap K) \trianglelefteq K^*(H\cap K)$

(3) $\dfrac{H^*(H\cap K)}{H^*(H\cap K^*)} \cong \dfrac{H\cap K}{(H^*\cap K)(H\cap K^*)} \cong \dfrac{K^*(H\cap K)}{K^*(H^*\cap K)}$.

pf: (1). By symmetry, it suffices to prove (1) & (3)

first, we show that $H^*(H\cap K)$ is a ~~group~~ subgroup of $G$.

Indeed, $H^* \trianglelefteq H$ and $H\cap K \leq H$. thus $H^*(H\cap K) \leq H \leq G$.

Next, we observe that $K^* \trianglelefteq K$. implies that $(H\cap K^*) \trianglelefteq (H\cap K)$

take ~~$a\in A_{i-1}, b\in A_i\cap B_{j-1}$~~. $h^*\in H^*$, $k^*\in H\cap K^*$, $h\in H^*$. $k\in H\cap K$.

then $hk(h^*k^*)(hk)^{-1} = h\underbrace{(kh^*k^{-1})}_{\in H^*}\underbrace{(kk^*k^{-1})}_{\in K^*\cap H}h^{-1} \in H^*(H\cap K^*)H^* = H^*(H\cap K^*)$ □.

(3). just prove: $\dfrac{H^*(H\cap K)}{H^*(H\cap K^*)} \cong \dfrac{H\cap K}{(H^*\cap K)(H\cap K^*)}$

Define a homomorphism $\phi$: $H\cap K \longrightarrow H^*(H\cap K) \longrightarrow \dfrac{H^*(H\cap K)}{H^*(H\cap K^*)}$.

$h \longmapsto h \longmapsto hH^*(H\cap K^*)$

$\ker\phi = \{h\in H\cap K \mid hH^*(H\cap K^*) = H^*(H\cap K^*)\}$

$= (H\cap K)\cap(H^*(H\cap K^*)) = (H^*\cap K)(H\cap K^*)$.

thus by first isomorphism theorem. we deduce.

$\dfrac{H\cap K}{(H^*\cap K)(H\cap K^*)} \cong \dfrac{H^*(H\cap K)}{H^*(H\cap K^*)}$ □.

Corallary. (Jordan–Hölder Thm). Suppose that $\{1\} = A_0 \triangleleft A_1 \triangleleft \cdots \triangleleft A_m = G$

$\{1\} = B_0 \triangleleft B_1 \triangleleft \cdots \triangleleft B_n = G$

then $m=n$. there is a bijection $\sigma: \{1,\dots,m\} \to \{1,\dots,n\}$. s.t. $A_{\sigma(i)}/A_{\sigma(i)-1} \cong B_i/B_{i-1}$

pf (sketched): $\dfrac{A_{i-1}(A_i\cap B_j)}{A_{i-1}(A_i\cap B_{j-1})} \cong \dfrac{(A_i\cap B_j)B_{j-1}}{(A_{i-1}\cap B_j)B_{j-1}}$

Some Group isomorphism Thm.

(1) $\phi: G \to H$ surjective homomorphism. then $H \cong G/\ker\phi$

(2) $A \leq G$, $B \trianglelefteq G$. Then $AB \leq G$.; $B \trianglelefteq AB$; $A \cap B \trianglelefteq A$.

Thus $\frac{AB}{B} \cong \frac{A}{A \cap B}$

(3) $H \trianglelefteq G$.; $K \trianglelefteq G$; $H \leq K$; then $\frac{K}{H} \trianglelefteq \frac{G}{H}$ and $\frac{G/H}{K/H} \cong \frac{G}{K}$

(4) (universal property) $N \trianglelefteq G$. then there is a bijection

$$\{H \mid N \leq H \leq G\} \longleftrightarrow \{H' \mid H' \leq G/N\}$$

preserves: intersections; normality; quotients of subgroups.

roducts, Group actions; Semi-products.

Direct product.

Thm 1.1. $G$ group. $H < G$. $K < G$.

(1) $H, G \trianglelefteq G$.

(2) $H \cap K = \{1\}$.

Then $HK \cong H \times K$ (there "$\times$" can be written as: "$\otimes$")

pf: (sketched) Define: $\phi: H \times K \to HK$ $(h,k) \to hk$.

Examples: 1. $m, n \in \mathbb{N}_+$. then $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$ iff $\gcd(m,n) = 1$.

2. Klein groups: $(K_4) V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

3. Fundamental thm of finitely generated Abelian groups:

$G$: finitely generated; Abelian, then

$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_s}$. $r \geq 0$, $n_i \mid n_{i+1}$

② Semi-direct product

Two definitions: ① $N, H$ groups. let $\phi: H \to \mathrm{Aut}(N)$ be a homomorphism.

then $N \rtimes H := \{(n,h) \mid n \in N, h \in H\}$.

$(n_1, h_1)(n_2, h_2) = (n_1 \cdot \phi_{h_1}(n_2), h_1 h_2)$

Thm ②. $G$ group. $N \trianglelefteq G$. $H \leq G$. suppose $H \cap G = \{1\}$.

Then $NH \leq G$ and $NH \cong N \rtimes H$.

Examples: ① $\mathbb{Z}_n \rtimes \mathbb{Z}_n^\times \cong \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \in M_{2\times 2}(\mathbb{Z}_n) \mid a \in \mathbb{Z}_n^\times \right\}$; ② $\mathbb{Z}_n \rtimes \{\pm 1\} \xrightarrow{\cong} D_{2n}$; ③

2. Group action:

Definition 2.1: $G$ group, $X$ set. A left $G$-action on $X$ is a map:

$G \times X \to X$ $(g, x) \mapsto g \cdot x$.

satisfying 1) $\forall x \in X$, $e \cdot x = x$.

2) $\forall g, h \in G$. $x \in X$, $g \cdot (h \cdot x) = (gh) \cdot x$

Remark. Given a group action of $G$ on $X$ $\Longleftrightarrow$ give a homomorphism of $S_X$.

Examples: 1. $S_n$ on $X = \{1, 2, \ldots, n\}$.

2. $D_{2n}$ on a regular $n$-gon.

$D_{2n} = \langle r, s \mid r^n = s^2 = 1, srs = r^{-1} \rangle$.

3. ①left translation action: $G \times G \to G$

$l_g: (g, x) \mapsto gx$.

②right ~: $G \times G \to G$

$(g, x) \to xg^{-1}$

③ Conjugation action: $G \times G \to G$

$Ad_g: (g, x) \mapsto gxg^{-1}$

By Remark. We conclude that: Every group $\cong$ subgroup of $S_G$.

(pf. Consider left translation action).

(1) An action is faithful iff. $g\cdot x = x$, $\forall x\in X \Rightarrow g=e$.

(2) An action is trivial iff $g\cdot x = x$, $\forall x\in X$.

(3) An action is transitive iff $\forall x, y\in X$. $\exists g\in G$. s.t. $x = gy$

Def 2.3. $G$ acts on $X$. $\forall\, x\in X$.

① Stabilizer group at $x$ : $= Stab_G(x) = \{g\in G \mid gx = x\}$.

② orbit of $x$ : $= Orb_G(x) = \{g\cdot x \mid g\in G\} \subseteq X$

Remark ① $Stab_G(x) \leq G$.

② $X = \bigsqcup_{\text{orbits } O} O$

③ $y\in Orb_G(x)$, then $Stab_G(y) = g\, Stab_G(x)\, g^{-1}$

Examples: Consider the conjugation action, of $G$. the orbits are called conjugacy classes.

① $G = GL_n(\mathbb{C})$. the representer of one orbits is a Jordan block.

② $G = S_n$, $n = \lambda_1 + \dots + \lambda_n$. $\lambda_1 \geq \dots \geq \lambda_n \geq 1$. $\longleftrightarrow$ one conjugacy classes.
where we use $\tau(a_1 a_2 \dots a_r)\tau^{-1} = (\tau(a_1)\ \tau(a_2) \dots \tau(a_r))$.

③ $G = D_{2n}$, ???

Def 2.4. $G$ group, $H\leq G$, $S\subseteq G$.

(1) $C_G(S) := \{g\in G \mid \forall s\in S, gs = sg\}$ is called the centralizer of $S$

(2) $Z(G) := \{g\in G \mid \forall h\in G, gh = hg\}$ is called the center of $G$.

(3) $N_G(H) := \{g\in G \mid gHg^{-1} = H\}$. is called the normalizer of $H$.

Remark. (1) Consider the conjugation action on $G$.
then $Stab_G(g) = C_G(g)$

(2) Consider the conjugation action on $\mathcal{P}(G)$
then $Stab_G(H) = N_G(H)$.

(3) $H \trianglelefteq N_G(H)$.

Def 2.5. $G$-equivariant : $G$ acts on $X$ and $Y$. then $\phi$ is $\sim$ iff
for $\forall g\in G, x\in X$. we have $\phi(gx) = g\cdot\phi(x)$

Prop. 2.6. $G$ is a transitive action on $X$. then there is a $\underline{G\text{-equivariant}}$ ① $\underline{\text{bijection}}$ ②.

$$\phi : G/H \xrightarrow{\cong} X$$
$$gH \longmapsto gx$$

Corollary 2.7. $G$ acts on $X$. then $X \cong \bigsqcup_{\text{orbits } O} O = \bigsqcup G/Stab_G(x)$

(Since $Orb_G(x) \cong G/Stab_G(x)$)

$G$ finite group. (action is conjugation).

(1) $\forall g \in G$. the number of elements in conjugacy class is

$$|Ad_G(g)| = |G| / |C_G(g)| = [G : C_G(g)]$$

(2). $g_1, \dots, g_r$ are representatives of conjugacy classes of $G$. $g_i \notin Z(G)$.

then $|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)]$

Examples: let $G = S_5$, the following are Stablizers

| | | | | |
|---|---|---|---|---|
| $S_5$ | $1+1+\dots+1$ | $S_2 \times Z_3$. | $2+3$ | $Z_2$. |
| $S_2 \times S_3$ | $1+1+1+1+2$ | $Z_2^2 \rtimes Z_2$ | $1+2+2$ | $Z_5$ 5 |
| $Z_3 \times S_2$ | $1+1+3$ | $Z_4$. | $1+4$ | |

Application: a structured proof of Sylow theorem.

Def 2.9. $Inn(G) := \{ Ad_g \in Aut(G) \mid g \in G \}$. , $Out(G) := Aut(G) / Inn(G)$

Example: let $G = GL_n(\mathbb{Q})$. $\Rightarrow \ker(Ad) = Z(GL_n(\mathbb{Q}))$

$Ad: G \to Aut(G)$. by linear algebra $\{ a \cdot I_n \mid a \in \mathbb{Q}^\times \} \cong \mathbb{Q}^\times$

Then $Inn(G) \cong GL_n(\mathbb{Q}) / \mathbb{Q}^\times := PGL_n(\mathbb{Q})$ (the projective general linear group)

Def $\psi : GL_n(\mathbb{Q}) \to GL_n(\mathbb{Q})$ $A \mapsto (A^{-1})^T$

Then we have a theory: $Aut(SL_n(\mathbb{Q})) \cong Aut(PGL_n(\mathbb{Q})) \cong \begin{cases} PGL_n(\mathbb{Q}) \rtimes \{1, \psi\} & n \geq 3 \\ PGL_2(\mathbb{Q}) & n = 2 \end{cases}$

where $SL_n(\mathbb{Q}) := \{ A \in GL_n(\mathbb{Q}) \mid \det A = 1 \}$.

Remark. (Dieudonné – O'Meara Thm). the structure of Automorphism of linear groups

let $K$ be a field. let $n \geq 3$. $\mathrm{char}(K) = 0$.

then $Aut(SL_n(K)) \cong PGL_n(K) \rtimes (Aut(K) \times \langle \psi \rangle) \cong Aut(PGL_n(K))$.

# Third proof of Sylow's theorems

Thm. Sylow's theorem. $|G| = p^r m$, $p \nmid m$

(1) Sylow p-subgroups exists

(2) All sylow p-subgroups are conjugate;

All p-power subgroups is contoined in a Sylow-p group

(3) Define $Syl_p(G) := \{$ Sylow-p-subgroups in $G\}$. $n_p = |Syl_p(G)|$

then 1.) $n_p \equiv 1 \pmod p$

2.) $n_p \mid m$

proof. (1) Suppose (1) is proved for order $< n$.

Let $G$ be the group of order $n = p^r \cdot m$. $p \nmid m$

1.). $r = 0$. then $\{1\} \trianglelefteq G$.

2). $p \nmid |Z(G)|$ then $|G| = |Z(G)| + \sum_{i=1}^{t} [G : C_G(g_i)]$

there exists $1 \le j \le t$. such that $p \nmid G/C_G(g_j)$.

thus $v_p(|C_G(g_j)|) = r$, by inductive hopothesis,

there exists $H$, where $|H| = p^r$. $H \le C_G(g_j) \le G$.

3) $p \mid |Z(G)|$, then $Z(G)$ is a finitely generated abelian group

$\Rightarrow Z(G) = \mathbb{Z}_{p^{r_1}} \times \cdots \times \mathbb{Z}_{p^{r_s}} \times \cdots =: \mathbb{Z}_{g(G)_p} \times \cdots$

$|Z(G)_p| = p^{r'}$, $(r' \ge 1)$

Consider $G \xrightarrow{\pi} G/Z(G)_p := \overline{G}$ $|\overline{G}| = p^{r-r'} \cdot m < n$

then there exists $H$, where $|\overline{H}| = p^{r-r'}$. $\overline{H} \le \overline{G}$

thus $H = \pi^{-1}(\overline{H})$, $|H| = p^r$. $H \le G$ □.

(2) let $P \le G$ is a sylow-p subgroup. $Q \le G$. $|Q| = p^{r'}$.

Consider $Q \curvearrowright G/P = \{gP \mid g \in G\}$.

$q \cdot gP := qgP$.

Then $|G/P| = \sum_{i=1}^{t} |Orbits_i| = \sum_{i=1}^{t} |Q/Stab_i|$ is not divisible by $p$.

thus there exists $Q' := \{q \in Q \mid gP = q \cdot gP\} = Stab_Q(gP) \le Q$

$v_p(|Q'|) = r'$. thus. $Q' = Q$.

thus $\forall\, q \in Q$, we have $qgP = gP \Rightarrow g^{-1}qg \in P$

$\Rightarrow Q \le gPg^{-1}$ □.

Corollary. 1). All sylow subgroups are conjugate.

2). There's only one Sylow-p-subgroup iff one sylow-p group is normal.

3). $N_G(N_G(P)) = N_G(P)$. and $N_G(P)$ contains a unique Sylow-p subgroup $P$

1)

(3) by (2) we say that the conjugation $G$-action on $Syl_p(G)$ is transitive

thus $n_p = |Syl_p(G)| = |G/Stab_G(P)| = |G/N_G(P)| = \dfrac{p^r \cdot m}{p^r [N_G(P):P]}$

where $P$ is a sylow-$p$ subgroup

$$|N_G(P)| = |N_G(P)/P|\,|P|$$

thus $n_p \mid m$

2.). Consider $P$ acting on $Syl_p(G)$ by conjugation. (this action is not transitive!).

$$n_p = \sum_{i=1}^{t} |P/Stab_P(P_i)|$$

1°. $Stab_P(P_i) \neq P$. then $Stab_P(P_i) \leq P$. then $p \mid |P/Stab_P(P_i)|$

2°. $Stab_P(P_i) = P$. then $P \subseteq N_G(P_i) \Rightarrow P = P_i$

then $n_p \equiv 1 \pmod p$ □.

Some easy applications:

the Structure of $G$ which has the order of $pq$. $p \neq q$. $p, q$ primes.

Thm. $p < q$. $|G| = pq$.

① $q \not\equiv 1 \pmod p$ : $G \cong Z_{pq}$.

② $q \equiv 1 \pmod p$ : 1. $G \cong Z_{pq}$; 2. $G \cong F_{p,q} := \langle a, b \mid a^q = b^p = 1,\ bab^{-1} = a^k \rangle$

where $k^p \equiv 1 \pmod q$

pf: (skipped).

The structer of $Z_q \rtimes_\eta Z_p$ $\quad \eta: Z_p \to Aut(Z_q) \cong Z_{q-1}^{\times}$

$(p \mid q-1)$

oups and Nilpotent groups

(1) The commutator of $x$ and $y$ : $[x,y] = x^{-1}y^{-1}xy$

(2) The commutator group of $G$: $G^{der} = G' = [G,G] = \langle [x,y] \mid x,y \in G \rangle$

Remark. (1) $xy = yx$ iff $[x,y] = 1$

(2) $g[x,y]g^{-1} = [gxg^{-1}, gyg^{-1}]$

(3) $G' \trianglelefteq G$, $G/G'$ is Abelian (which is the motivation of the definition)

pf. (3) (sketched) $\forall\ \bar{x} \neq \bar{y} \in G/G'$. $[x,y] \in G' \Rightarrow [\bar{x},\bar{y}] = \bar{1} \Rightarrow \bar{x}\bar{y} = \bar{y}\bar{x}$.

Def 1.2. Solvable groups : a group $G$ is called solvable iff there exists a chain: $\{1\} = G_0 \trianglelefteq G_1 \trianglelefteq \dots \trianglelefteq G_r = G$

satisfying 1) $G_{i-1} \trianglelefteq G_i$ ; 2) $G_i/G_{i-1}$ is Abelian.

Def 1.3. For any group $G$, define the following sequence:
$G^{(0)} = G$, $G^{(1)} = [G,G]$, $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. is called the commutator series of $G$.

Prop. 1.4. A group is solvable iff $G^{(n)} = \{1\}$ for $n < \infty$

pf: "$\Leftarrow$" is trivial;

"$\Rightarrow$": $\{1\} = H_0 \trianglelefteq H_1 \trianglelefteq \dots \trianglelefteq H_r = G$. ; $G_i/G_{i-1}$ is Abelian.

thus $[H_{i-1}, H_{i-1}] \subseteq H_{i-1}$. ( $G/N$ Abelian $\Leftrightarrow$ $\ker \pi \supseteq [G,G]$ ).

thus. $[G,G] \leq H_{r-1}$

$G^{(2)} \leq [H_{r-1}, H_{r-1}] \leq H_{r-2} \Rightarrow \dots \Rightarrow G^{(r)} \leq H_0 = \{1\}$. □.

Def 1.5. Characteristic groups: $G$ a group, $H \leq G$ is characteristic iff

for any automorphism $\sigma$ of $G$, $\sigma(H) = H$. noted by $H \text{ char } G$

Prop. 1.6. (1) Characteristic groups are normal.

(2) All $G^{(i)}$ are characteristic groups of $G$.

(3) $H \leq G$ then $H^{(i)} \leq G^{(i)}$, thus $G$ is solvable $\Rightarrow$ $H$ is solvable.

(4) $\phi: G \to K$ surjective, homomorphism. then $\phi(G^{(i)}) = K^{(i)}$

thus $G$ solvable $\Rightarrow$ $K$ solvable.

(5) $N \trianglelefteq G$ and $N$, $G/N$ are solvable, then $G$ solvable.

pf: (trivial).

Def 1.7. (1) $G$ a group define the sequence

$G^0 = G$, $G^1 = [G,G]$, $G^{(i)} = [G, G^i]$ is called the lower central series of $G$.

(2) $G$ is called nilpotent iff $G^c = \{1\}$ for $c \in \mathbb{N}$.

$c$ : nilpotence class of $G$.

Remark. 1) $G^i \trianglelefteq G$. ; $G^{(i)} \leq G^i$.

2) let $N \trianglelefteq G$ ; $\pi : G \to G/N = H$ then $\pi(G^i) = H^i$.

3). Nilpotent groups are solvable.

Def 1.8. $G$ group. define the sequence.

$Z_0(G) = 1$. $Z_1(G) = Z(G)$. $Z_2(G) = \pi^{-1}(Z(G/Z(G)))$

$\cdots$ $Z_{i+1}(G) = \pi^{-1}(Z(G/Z_i(G)))$.

$$\begin{array}{ccc} G & \xrightarrow{\pi} & G/Z_i(G) \\ \cup & & \cup \\ Z_{i+1}(G) & \longrightarrow & Z(G/Z_i(G)) \end{array}$$

that is $Z_{i+1}(G)/Z_i(G) \cong Z(G/Z_i(G))$.

$\{1\} = Z_0(G) \trianglelefteq Z_1(G) \trianglelefteq \cdots$ is called the upper central series of $G$.

Thm 1.9. A group $G$ in nilpotent iff $G = Z_c(G)$ for some $c \in \mathbb{N}$.

pf: Induction on the minimal $c$ such that either $G^c = \{1\}$ or $Z_c(G) = Z(G)$.

$c = 1$. $Z(G) = G \iff G$ Abelian $\iff [G,G] = G^1 = \{1\}$

for all $< c$ the proposition is correct.

Let $\pi : G \longrightarrow G/Z(G) := \overline{G}$.

we will prove: $G^c = \{1\}$ $\quad$ $Z_c(G) = G$.

$\Updownarrow$ (1) $\qquad$ $\Updownarrow$ (2)

$(\overline{G})^{c-1} = \{1\}$ $\overset{\text{inductive hypothesis}}{\Longleftarrow}$ $Z_{c-1}(\overline{G}) = \overline{G}$

(1). $G^c = \{1\} \Leftrightarrow [G, G^{c-1}] = \{1\} \Leftrightarrow G^{c-1} \leq Z(G) \Leftrightarrow (\overline{G})^{c-1} = \{1\}$.

(2).

$$\begin{array}{ccccccc} G & \supseteq & Z_1(G) & \hookrightarrow & Z_2(G) & \hookrightarrow & Z_3(G) \\ \downarrow \pi_1 & & & & \downarrow \pi_1 & & \downarrow \pi_2 \\ G/Z_1(G) = \overline{G} & & \supseteq & & Z_1(\overline{G}) & \hookrightarrow & Z_2(\overline{G}) \\ & \searrow^{\pi_2} \quad \downarrow \overline{\pi}_1 & & & & & \downarrow \overline{\pi}_1 \\ G/Z_2(G) = \overline{G}/Z_1(\overline{G}) & & & & & & Z(\overline{G}/Z_1(\overline{G})) \end{array}$$

thus $\dfrac{G}{Z_2(G)} \cong \dfrac{G/Z_1(G)}{Z_2(G)/Z_1(G)} = \dfrac{\overline{G}}{Z_1(\overline{G})}$. (*)

$\Rightarrow Z(G/Z_2(G)) \cong Z(\overline{G}/Z_1(\overline{G}))$.

At the same time, $Z_2(\overline{G}) = \overline{\pi}_1^{-1}(Z(\overline{G}/Z_1(\overline{G})))$ (**)

$\Rightarrow Z_3(G) = \pi_2^{-1}(Z(G/Z_2(G))) \overset{*}{=\!=} \pi_1^{-1}(Z(\overline{G}/Z_1(\overline{G}))) \overset{**}{=\!=} \pi_1^{-1}(Z_2(\overline{G}))$.

thus: $Z_c(G) = G \iff Z_{c-1}(\bar{G}) = \bar{G}$. □

Remark. More precisely, we have $G^{c-i} \le Z_i(G)$

Since by inductive hypothesis, $\bar{G}^{c-i} \le Z_{i-1}(\bar{G})$

$\Rightarrow G^{c-i} \le \pi_i^{-1}(\bar{G}^{c-i}) \le \pi_i^{-1}(Z_{i-1}(\bar{G})) = Z_i(G)$

Thm 1.10 (Structure thm of nilpotent groups) let $|G| = n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ $P_i \in Syl_{p_i}(G)$

Then TFAE: (1) $G$ is nilpotent;

(2) $H \lneq G \Rightarrow H \lneq N_G(H)$.

(3) All sylow subgroups $P_i$ are normal

(4) $G \cong P_1 \times P_2 \times \cdots \times P_r$

Lemma 1.11. All p-groups are nilpotent.

pf: $|P| = p^\alpha$, $(\alpha \ge 1)$. then $p^\alpha = |Z(p)| + \sum_{i=1}^{r} |P : C_P(h_i)|$

↳ nontrivial p-power

$\Rightarrow p \mid |Z(p)|$, $\{e\} \le Z(P)$ thus $Z(P)$ is nontrivial □.

Lemma 1.12. Let $P$ be a p-group.

(1) $Z(P) \neq \{e\}$.

(2) If $H \trianglelefteq P$. $H$ nontrivial. then $H \cap Z(P) \neq \{e\}$

(3) If $H \lneq P$. then $H \lneq N_P(H)$.

pf: (1) ✓; (2) Consider the conjugation action of $P$ on $H$:

then $H = \coprod_i P/Stab_P(a_i)$ where $Stab_P(a_i) = P \iff a_i \in Z(P) \cap H$

$\Rightarrow 0 \equiv |H| \equiv \sum_{i=1}^{r} |P|/|Stab_P(a_i)| = |Z(P) \cap H| \pmod p$

Since $H$ is nontrivial, thus $H \cap Z(P) \neq \{e\}$

(3) ① $Z(P) \nsubseteq H$; $Z(P) \trianglelefteq P \Rightarrow H \lneq N_P(H)$.

② $Z(P) \subseteq H$; consider $\bar{H} = H/Z(P)$. $\bar{P} = P/Z(P)$, By induction.

$\bar{H} \lneq N_{\bar{P}}(\bar{H})$. thus $H \lneq N_P(H)$. □.

Back to the proof:

(1) $\Rightarrow$ (2) ① $Z(G) \nsubseteq H$, ...

② $Z(G) \le H$ consider $\bar{H} = H/Z(P)$, $\bar{P} = P/Z(P)$ ...

(2) $\Rightarrow$ (3) Note that $N_G(N_G(P_i)) = N_G(P_i)$. ($N_G(P_i) \trianglelefteq N_G(N_G(P_i))$; $N_G(N_G(P_i)) \trianglelefteq N_G(P_i)$)

if $N_G(P_i) \lneq G \Rightarrow N_G(P_i) \lneq N_G(N_G(P_i)) = N_G(P_i)$ contradiction.

(3) $\Rightarrow$ (4) $P_i \cap P_j = \{e\}$. $(i \neq j)$. $P_i$ (normal) $\Rightarrow G \cong P_1 \times \cdots \times P_r$.

(4) $\Rightarrow$ (1): $P_i$ is nilpotent. then $G = \prod_{i=1}^{n} P_i$

$\Rightarrow Z^c(G) = \prod_{i=1}^{n} Z^c(P_i)$. take $c = \max\{c_1, \ldots, c_n\} \Rightarrow Z^{(c)}(G) = G$ □

A general and brief introduction of the ring theory.

Def 2.1. (1) $R$ is a ring iff (1) $(R,+)$ is an Abelian group

(2) $(R,\cdot)$ is a monoid.

(3). $1 \neq 0$

(4) for $a,b,c \in R$ $\quad a(b+c) = a\cdot b + a\cdot c$ ; $a\cdot(b+c) = ab + a\cdot c$.

(2) $R$ is a division ring iff $\forall x \in R$, $\exists\, x^{-1} \in R$. s.t. $x \cdot x^{-1} = x^{-1} \cdot x = e$

(3) $R$ is a field iff $R$ is a commutative division ring.

(4) $R$ is an integral domain iff $R$ is a commutative ring with no zero-divisor.

In the following lecture, we just take $R$ commutative.

Remark. A finite integral domain $R$ is a field

Pf: (sketched) Define $\phi_a : (R,+) \longrightarrow (R,+)$, $x \longmapsto ax$

Def 2.2 A left ideal is a subset $I \subseteq R$ iff (1) $a-b \in I$. $\forall a,b \in I$.

(2) $xa \in I$. $\forall x \in R$. $a \in I$.

Notation: define the ideal generated by $a_1, \dots, a_s$ : $(a_1, \dots, a_s) = \left\{ \sum_{i=1}^{s} x_i a_i \mid x_i \in R \right\} \subseteq R$

Def 2.3. $R$ ring, $I$ ideal then the quotient ring: $R/I := \{ x+I \mid x \in R \}$

with operations: $x+I + y+I = (x+y)+I$. $\quad (x+I)(y+I) = (xy)+I$

Examples: $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$.

Thm 2.4. (1) $\phi : R \to S$. surjective, homo~ $\Rightarrow$ $S \cong R/\ker\phi$.

(2) $I \subseteq J$ ideal of $R$ then $(R/I)/(J/I) \cong R/J$.

Def 2.5. The fraction field of the integral domain $R$

$$\mathrm{Frac}(R) := \{(a,b) \in R \times (R \setminus \{0\})\} / \sim .$$

where $(a,b) \sim (c,d)$ iff $ad = cb$. hence $\mathrm{Frac}(R)$ is a field

Examples. $\mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}$. $\quad \mathrm{Frac}(k[x]) = k(x)$

(also remainder thm)

Let $I_1, \dots, I_k$ be ideals of $R$. $I_1, \dots, I_k$ are pairwise comaximal

(1) $\phi: R \longrightarrow R/I_1 \times \cdots \times R/I_k$ is surjective;

(2) $I_1 \cap \cdots \cap I_k = I_1 \cdots I_k$

(3) $R/I_1 \cdots I_k \cong R/I_1 \cap \cdots \cap I_k \cong \frac{R}{I_1} \times \cdots \times \frac{R}{I_k}$

pf (skipped)

Corollary 2.7. $n_1, \dots, n_r$ are pair-wise coprime. then $\frac{\mathbb{Z}}{n_1\mathbb{Z}} \times \cdots \times \frac{\mathbb{Z}}{n_r\mathbb{Z}} \cong \frac{\mathbb{Z}}{n_1 n_2 \cdots n_r \mathbb{Z}}$

Def 2.8. Some special ideals.

(1). $m \subseteq R$ is called maximal if $m \neq R$ and the only ideals containing $m$ are $m$ & $R$

(2) A proper ideal $\mathfrak{p} \subsetneq R$ is called a prime ideal if

$ab \in \mathfrak{p} \Rightarrow a \in \mathfrak{p}$ or $b \in \mathfrak{p}$

Prop. 2.9. (1) $m \subseteq R$ is maximal iff $R/m$ is a field.

(2) $\mathfrak{p} \subseteq R$ is a prime ideal iff $R/\mathfrak{p}$ is an integral domain

(3) maximal ideal $\Rightarrow$ prime ideal.

pf: consider the $\phi: m \to R/m$ & the lattice isomorphism thm. (sketched).

Remark. Prop. 10.4.5 is a good exercise.

Def. 2.9. Let $R$ be an integral domain.

(1). $R$ is said to be a UFD. if $\forall r \in R$. $\exists!$ $p_1, \dots, p_n$ irreducible (under the sense of association), s.t. $r = p_1 p_2 \cdots p_n$.

(2) $R$ is said to be a PID if every ideal is principle.

(3) $R$ is said to be a ED if there exists $Nm: R \to \mathbb{Z}^+ \cup \{0\}$. s.t.

① $Nm(0) = 0$

② $\forall a, b \in R$. with $b \neq 0$. $\exists q \in R$. $r \in R$. s.t. $a = bq + r$. $0 \leq Nm(r) < Nm(b)$.

Some useful & basic properties.

Prop. 2.10. (1) $R$ is UFD. then prime elements $\Leftrightarrow$ irreducible elements.

(2) $R$ is PID then maximal ideal $\Leftrightarrow$ prime ideal.

(3) $F$ is a field then $F[x]$ is an ED

Thm 2.11. ED $\overset{(1)}{\Rightarrow}$ PID $\overset{(2)}{\Rightarrow}$ UFD.

pf. (1) $\Rightarrow$ (2) Consider $b = aq + r$. $\deg Nm(r) < Nm(a)$.
(sketched)

(2) $\Rightarrow$ (3) Consider $r \in R$. $r = a_1 \cdot b_1 = a_1 a_2 b_2 = a_1 a_2 a_3 b_3 = \dots$

$\Rightarrow (r) \subseteq (b_1) \subseteq \dots$

Then take $I = \bigcup_{n \geq 0} (b_n) : \overset{\text{PID}}{=\!=} (b)$

Uniqueness: $r = p_1 \cdots p_n = q_1 \cdots q_m$ and use induction $\square$.

Thm 2.12. R is an integral domain, then R is a UFD $\Leftrightarrow$ R[x] is a UFD.

pf (Sketched): $\Rightarrow$ Using Gauss Lemma: take $F = \text{Frac}(R)$. $p(x) \in R[x]$ is reducible in $F[x]$ then $p(x)$ is reducible in $R[x]$ $\square$

A useful lemma 2.13 (in the Field theory): F be a field. $f(x)$ is irreducible iff $F[x]/(f(x))$ is a field.

pf: $f(x)$ irreducible $\overset{\text{UFD}}{\Leftrightarrow}$ $f(x)$ prime elements

$\overset{\text{PID}}{\Leftrightarrow}$ $F[x]$ $(f(x))$ maximal in $F[x]$.

$\Leftrightarrow$ $R/(f(x))$ is a field $\square$.